Veritas Document 3C

FEAT Transparency Principles Assessment Methodology

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Executive Summary

With many decisions in the Financial Services Institutions (FSIs) now being shaped or even made by algorithms, there is a greater demand for transparency around crucial questions (e.g., where are algorithms in use? what data do they use? how are they making decisions? what impact do they have?

Regulators around the world are encouraging FSIs to increase transparency around Artificial Intelligence and Data Analytics (AIDA) decisions for people who are impacted by such decisions. This includes three of the 14 Principles (see section 1.2 for the list of Principles) under the Monetary Authority of Singapore's Fairness, Ethics, Accountability and Transparency (FEAT) guidelines released in 2018.

FSIs can choose to see this demand solely from the perspective of potential future regulatory compliance requirements. However, it should arguably be a core business priority in its own right. In an increasingly digital world, transparency is an unmet customer need, and providing greater transparency could help FSIs build stronger customer relationships and create competitive advantage. In addition, creating greater internal transparency into such algorithms can help overcome the trust deficit that AIDA systems may suffer within the FSI and/or with regulators.

Fundamentally, algorithms such as those built using advanced Machine Learning (ML) techniques learn from the data used to train them. As a result, they are more susceptible to failing when the real world data changes significantly or when they inadvertently accentuate existing biases in society.

FSIs' approach to AIDA transparency need to address two aspects:

- **External.** What level and form of transparency should FSIs provide to their customers/ prospects and other external stakeholders in various situations?
- **Internal.** How should FSIs go about building reliable explanations to feed into (and provide verifiable backup for) the information shared externally and to be compliant with relevant internal policies and procedures and regulatory requirements?

With respect to external transparency, a variety of regulatory considerations apply, such as existing ones around treating customers and staff fairly and respecting an individual's personal data rights and new ones specific to AIDA systems. As a result, it is difficult to prescribe the level of transparency needed for different AIDA use cases.

However, it is possible to provide a checklist of the key considerations for an FSI in arriving at such a decision. The materiality of the AIDA-driven decision to the impacted data subject (e.g., whether it results in denial of access to financial service) need

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to be weighed against considerations such as the prevention of fraud or gaming by unscrupulous applicants or the FSI's legal obligations such as financial crime prevention/ detection.

There are also practical considerations to be kept in mind in the operationalisation of transparency. For example, human beings are limited in their ability to interpret complex explanations for algorithmic decisions. The reasons behind a decision may be helpful in educating them, but sometimes it may be more valuable to highlight the actions they can take to change the outcome.

With regard to internal transparency, the past few years have seen significant activity around designing explanation methods for AIDA systems, particularly with respect to complex ML models. FSIs should review the existing landscape of methods and consider implementing internal standards for approved explanation methods.

More broadly, FSIs should take steps to implement transparency requirements in a scalable way, rather than on a case-by-case basis. Section 4 of this paper makes five recommendations in this regard: embedding AIDA transparency considerations into existing risk frameworks; investing in proactive communication; training customers, staff, management and risk/ control functions; investing in tools for AIDA transparency; and integrating transparency considerations throughout the AIDA lifecycle.

Finally, it should be recognised that the field of algorithmic transparency is a relatively immature one. There are multiple areas in which there is room for future work around regulatory guidance, industry practices and even academic research (e.g., the alignment of transparency considerations between AIDA and non-AIDA systems, or the treatment of third-party data and models with respect to transparency requirements).

01 Introduction

1.1 Scope, Purpose and Intended Audience of the Document

This document* is intended for Artificial Intelligence and Data Analytics (AIDA) practitioners, business users of AIDA systems and their risk/control counterparts in Financial Services Institutions (FSIs).

The document acknowledges that an FSI's specific obligations around AIDA transparency will depend on the regulatory requirements and societal norms of the jurisdictions in which they operate, and the materiality of individual use cases. The former set of expectations are also expected to change over time. As a result, the document does not attempt to prescribe what level of transparency should be provided to data subjects in specific AIDA use cases. Instead, it focuses on:

- Providing a methodology to evaluate the need for external transparency (i.e., facing the data subject impacted by AIDA decisions) and to operationalise it in a pragmatic way.
- Recommending a set of internal capabilities needed to underpin such transparency with data subjects and build AIDA trustworthiness within the FSI organisation.

The document also recognises the need to embed transparency considerations into an FSI's processes and systems in a scalable way, and records specific areas in which the industry will benefit from greater regulatory clarity and/or customer research.

This document is one of a suite published as a deliverable of the Monetary Authority of Singapore (MAS) Veritas Phase 2 project, and it fits alongside published documents as highlighted in the diagram below.

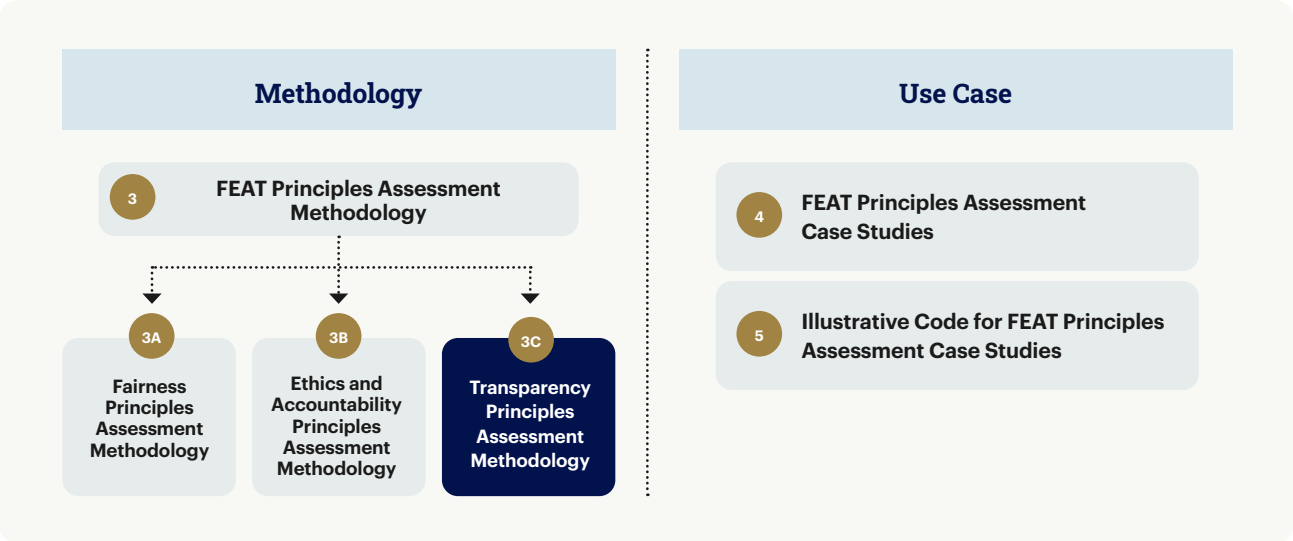


Figure 1.1 Transparency Principles Assessment Methodology document in the suite of Veritas outcomes

Audience of this document is aligned with the audience listed out in section 1.3 of Veritas Document 3 FEAT Principles Assessment Methodology document

1.2 Background - Regulatory context of Transparency in AIDA

As AIDA systems are increasingly used to shape decisions or engage customers in the FSI, regulators worldwide are encouraging FSIs to ensure sufficient transparency around their use. In the United States, reasons for denial must be provided to rejected (individual) customers in lending scenarios. Under the European Union's General Data Protection Regulation (GDPR), organisations that deploy automated decision making must provide the individuals affected with meaningful information about the logic behind decisions.

Empirical evidence confirms the importance of explanations to end users, and that explanations can significantly increase a users' confidence and trust[1,2,3] while also enhancing their ability to assess the correctness of a prediction.[4,5]

In Singapore, three of the Principles outlined in the MAS Fairness, Ethics, Accountability and Transparency (FEAT) guidelines (2018)[6] on the use of AIDA explicitly relate to transparency:

- **Principle 12:** To increase public confidence, use of AIDA is proactively disclosed to data subjects as part of general communication.
- **Principle 13:** Data subjects are provided, upon request, clear explanations on what data is used to make AIDA driven decisions about the data subject and how the data affects the decision.
- **Principle 14:** Data subjects are provided, upon request, clear explanations on the consequences that AIDA driven decisions may have on them.

In addition, transparency is essential for an FSI to adhere to other FEAT principles such as fairness and accountability. The Singapore Personal Data Protection Commission (PDPC) guidelines[7] highlight expectations around algorithmic explainability and how it can be meaningfully brought to life for consumers. Similar guidelines have been, or are being, issued by regulators elsewhere in the world.



1.3 Illustrative Examples of AIDA Transparency

FEAT Principle 12 suggests that FSIs proactively disclose to data subjects when AIDA systems are being used to engage with them (e.g., when interacting with an automated voice agent rather than a human) or make a decision affecting them (e.g., underwriting decisions being made by humans in tandem with an AIDA system).

FEAT Principle 13 suggests that where a data subject requests it, an FSI should provide:

- **A description of the data being used by an AIDA system.** And the associated human oversight involved in making decisions that affect them. In insurance, this could involve disclosing during the application process that the applicant's age and the results of their initial health assessment will be used in determining their life insurance premium. In banking, an example would be to disclose during a credit card application that the applicant's annual income, historical current account balance and bureau credit rating will be used to determine their credit card spending limit.
- **An explanation of the way in which the data affects the decision.** An insurance example would be to explain that pre-existing health conditions will drive up the premium, or that positive health indicators from the customer's personal health device, if shared with the insurer, will activate subsequent rebates. A banking example would be to share that the lack of adequate credit history has a negative impact on the customer's credit worthiness.

FEAT Principle 14 suggests that upon request, an FSI should explain to a data subject how AIDA driven decisions could affect them. For example, an FSI could let prospects and customers know that if the AIDA system being used to assess their insurance or loan application does not approve it, the application is not automatically rejected, but is instead forwarded to a human assessor. A further example would be to disclose to prospects and customers that the result of their current insurance or loan application will be used as an input when deciding on any future applications to the FSI within the next "X" months.

1.4 (Why) Do AIDA Driven Decisions Require Enhanced Transparency?

Transparency requirements are not unique to AIDA driven decisions. Existing requirements around treating customers and staff fairly, ensuring equal opportunities for different groups in society and respecting the rights of individuals around their personal data already cover similar ground.

Three factors have led to the increased attention on transparency in the specific context of AIDA driven decisions:

1. Some types of AIDA systems use “black box” algorithms.
2. Many of today’s AIDA systems use more data about data subjects and from more sources than their historical counterparts.
3. The impact of poor AIDA driven decisions can be more widespread and perpetuated at greater speed than historical manual or automated alternatives, creating a risk of systematic misuse.

However, it is difficult to isolate the transparency standards and processes for AIDA driven decisions from those used for other types of decisions. In many cases, AIDA systems are used to complement traditional decision making processes, rendering such distinctions impractical. For example, could an FSI explain to a prospect or customer why an AI algorithm recommended giving them a loan, but then not explain why a human approver overseeing the decision overruled it? Additionally, the perception of what constitutes an AIDA system might vary between, and even within, FSIs. Therefore, creating separate operating procedures for explaining AIDA and non-AIDA decisions may be impractical. One way of addressing this issue would be to focus transparency efforts on the end-to-end process used to arrive at a decision, rather than just the AIDA subcomponent.



1.5 Benefits of Transparency: Beyond Regulatory Compliance

FSIs could see the demand for greater transparency around AI purely from the perspective of potential future regulatory compliance requirements. However, this would represent a missed opportunity. FSIs should adopt greater transparency and ensure fair, ethical and accountable use of AIDA systems because it meets an unmet customer need and makes sense from a business perspective.

In most societies, the use of data and algorithms (whether AI enabled or otherwise) has moved significantly ahead of the ability of the general public to comprehend it. There have been well publicised examples – of varying levels of accuracy – of malfunctioning algorithms and their adverse impact on individuals. FSIs could potentially use greater transparency around the use of AIDA systems to increase customer trust, strengthen customer relationships and create a source of competitive advantage.

Another important benefit to FSIs from greater transparency is the contribution it can make toward overcoming internal barriers to the more widespread adoption of AIDA. Particularly when it comes to algorithms created using advanced Machine Learning (ML) techniques, trustworthiness concerns have held back adoption at scale. Concerns include worries that ML:

- May not always be explainable.
- Could be affected by a drop off in predictive accuracy in the face of rapid changes, such as those seen during the COVID-19 pandemic.
- Run the risk of accentuating biases based on the data used to train them.
- "Learn by memorisation" and draw false patterns from limited past data, a concept known as "overfitting".

In addition, by establishing the processes and systems that enable external transparency, FSIs will be able to create a robust internal basis to win over AIDA sceptics inside their organisations and capture its full value. Individuals within the FSI will also be able to build a better understanding of the AIDA system, as a prerequisite to taking accountability for its outcomes.

02 Key Transparency Concepts

2.1 External Transparency: Overview

In the information FSIs provide to data subjects on AIDA use, they can be proactive or reactive; generic or instance specific; and informational or action oriented:

- **Proactive vs. Reactive:** proactive transparency involves an FSI informing the general public, including current and prospective customers, that it uses their data and AIDA systems to engage them and/or make decisions that affect them. This type of transparency corresponds to FEAT Principle 12. Reactive (or responsive) transparency involves the FSI responding to requests from data subjects to find out more about a specific decision that has affected them (e.g., why did my loan application not get approved? Why am I being charged a higher insurance premium than my husband? How will this decision impact me? etc). This type of transparency corresponds to FEAT Principles 13 and 14.
- **Generic vs. Specific:** proactive transparency can be both generic and instance specific. Information provided on websites and on product marketing collateral (such as that for a new credit card) about the FSI's use of data and algorithms would be examples of the former. Information provided during the application process or in formal customer documentation would be examples of the latter. Reactive transparency is by definition instance specific (e.g., related to a particular loan or insurance application).
- **Informational vs. Action Oriented:** the purpose of providing transparency to data subjects can be informational (i.e., helping them understand how the FSI uses data and algorithms to engage them and make decisions impacting them). Alternatively, providing information can be action oriented (i.e., helping the data subject seek an alternative decision by taking specific steps, such as changing specific aspects of their application).

2.2 External Transparency: Content of Explanations

Early studies from cognitive psychology suggest how humans might view interpretations of AIDA systems. A suitable explanation for AIDA driven decisions is expected to meet three conditions:

1. It must be consistent with human intuition.
2. Its level of complexity must match the expertise level of the humans affected.
3. It must use domain specific (i.e., specific to the use case) vocabulary.

Research suggests that human interpretable explanations cannot be too lengthy, but that humans still require understanding of the core factors that underpin decision making.[9] Humans generally prefer such explanations to be simple and highly probable,[10] and to appeal to causal structure rather than correlations.[11,12] A natural progression – from broad explanations of how the system works to justifying decisions in a particular instance – may be appropriate in many instances.[13]

In addition, data subjects ideally require a mix of reasoning (general explanations for their decision) and action (an understanding of how they can change a model's behaviour). Adding counterfactuals (see details in section 2.4.4) that demonstrate how a decision could be improved or changed can help data subjects grasp model behaviour (e.g., "had I done X, would my outcome Y have changed?").

Counterfactuals have played a role in the philosophy of explanations dating back to 1973,[14,15] and psychological studies have confirmed that counterfactuals can elicit causal reasoning in humans.[16]

A thorough explanation would not only allow a data subject to understand the causes of a model's decision, but also help the user mentally simulate some aspects of the model itself.[17]

In applying these considerations, FSIs should ensure that they have answers to the following questions as part of their reactive explanations:

1. **How was the decision made?** A short overview of the overall decision making process, including the role of the AIDA system as well as any associated human oversight, and an overview of the data used to make the decision. The overview will be specific to this particular use case, but not to the data subject.
2. **What were the top reasons behind your specific decision?** The top few reasons that drove the decision for this particular data subject. Ideally, this includes not just the factors that acted against the data subject but also the ones that acted in their favour.
3. **What actions could have enabled a more favourable outcome for yourself?** Sharing actions that might help the data subject get a more favourable outcome can appear intuitively more attractive than just sharing the top reasons, since the latter may not always be actionable (e.g., annual income cannot be changed overnight, and age cannot be reduced at all!). However, it should be limited to instances in which the FSI can guarantee a (favourable) change in outcome if the data subject takes those actions. It should also be accompanied by appropriate time limitations (e.g., only valid for a month or three months after the original decision), since the underlying algorithm and/or manual processes may change over time. Finally, the FSI needs to take care to not recommend actions in instances where such recommendations would constitute an inappropriate influencing of the data subject's behaviour.
4. **How does the AIDA decision impact you?** Starting with the obvious (i.e., that you were denied the loan, or your insurance claim was rejected, as a result of the AIDA decision and/or associated human oversight). This summary could also relate to broader impact (e.g., that the result of this assessment will be used to determine the outcomes of your future applications in the next X months).
5. **What redress options are available to you?** Linked to FEAT Principles 10 and 11, to enable data subjects to enquire about, submit appeals for and request reviews of AIDA driven decisions, and to provide relevant supplementary data during review.

2.3 Internal Transparency: Overview and Taxonomy

While FSIs can decide on the appropriate level of transparency to be provided externally to data subjects impacted by an AIDA driven decision, they have less flexibility when it comes to internal transparency. Being able to understand and explain how the AIDA system is making its decision or recommendation, as well as the human oversight rules being used on top, is essential for two reasons:

1. To ensure the accuracy of explanations to data subjects.
2. To win the trust of internal stakeholders as a precondition for AIDA use in production. Some of the latter have to take personal accountability for AIDA driven decisions, and a robust explanation of the AIDA system's inner workings is critical for them.

Existing techniques already provide adequate explanations for some types of AIDA systems, such as rule based and traditional statistical models. This section focuses on the mechanisms to enable such explanations for ML AIDA systems, where the challenge around lack of transparency is at its most acute. ML explainability is a broad and often nebulous concept. A taxonomy of ML explanation methods can be built by answering four core questions:

1. **Does the model use specific or agnostic methods?** Model specific explanation methods (e.g., those used for looking at branching structures within trees or the weights learned by a layer of a neural network) are useful for a defined class of model and may require access to the internal structure of a model. On the other hand, model agnostic methods do not expect any access to the internal workings of the model. Agnostic methods might then create explanations only by examining model input and output pairs.
2. **What is the scope of the explanation?** Explanation methods are fundamentally characterised by the quantity it is attempting to explain. Is it an individual model prediction (a local explanation) or the overall aggregate drivers of model behaviour (global explanations)? Is it an internal component of the model? Is it a segment of the overall population?
3. **What is the output of the method?** Explanation methods must summarise their results with respect to a component of the model.
    a. In traditional explanation frameworks, it is usually features, where a method will attempt to explain the local or global properties of the model by explaining how each feature affected the overall model decision(s).
    b. For more complex models, explanations might instead target model internals such as learned weights or layers, such as the branching structure of a decision tree or visualisation of intermediate layers in a neural network.
    c. Still other explanation methods might explain model outputs with respect to the underlying training or test data points (i.e., the points that most contributed to a certain aspect of learned model behaviour, or the ways in which one might modify a data point to change a model's predictions).
4. **Is the explanation intrinsic to the model (i.e., is the model inherently interpretable)?** Explanation methods can follow naturally from the structure of the model (such as in the case of a decision tree), or be retroactively applied to understand a more opaque model (such as a neural network). Examples of inherently interpretable architectures include GLMs,[18] GAMs,[19] and simple trees.

2.4 Internal Transparency: Current State of ML Explanation Methods

This section focuses on explanation methods that specifically attribute model behaviour to constituent features or internal layers. It provides an overview of the explanation methods available today, organising them first by whether they are "baked" into the model architecture (inherently interpretable) or are post hoc in nature. Explanation methods are then characterised by their access to the underlying model (model specific vs. model agnostic) and their scope (local vs. global).

The final part of this section outlines other explanation techniques that attempt to interpret a model with respect to specific data points (via either attribution to training data[20] or by providing counterfactual points that would have changed a model outcome).[21] **Due to their technical nature, some parts of this section may be more suited to AIDA practitioners than their generalist counterparts.**

2.4.1 Post Hoc Interpretability: Model Agnostic Methods

Post hoc interpretability consists of explanations that are generated from a model outcome once it has already been trained. This approach differs from inherently interpretable models that are architected with some interpretability constraint in mind. This subsection deals with model agnostic explanation methods, while the next covers model specific methods.

Model agnostic interpretability methods offer increased flexibility in that they can be used with any ML technique as a post processing step. In general, a good model agnostic technique will work across a wide range of model classes and allow for interpretation of results in a different feature space than the potentially complex input space of the model.

- **Global Explanation Methods.** Early model agnostic methods such as **Partial Dependence Plots (PDPs)**[22] and **Accumulated Local Effects plots (ALE plots)**[23] focused on a global interpretation of a model's behaviour. PDPs and ALEs both capture the marginal effect of an individual feature or a small set of features on the average output of a model. PDPs in particular examine how a model's output changes if the value of a feature of influence is varied and calculates the expected model output over the marginal distribution of the unfixed features. Yet another similar technique is **Permutation Importance**,[24] which estimates a feature's importance by seeing how much the model's error increases if that feature's value is randomly permuted amongst all data points.[25] In this way, the permuted feature becomes (on average) uncorrelated with outcomes and therefore will not help in prediction.

  Other model agnostic global interpretation techniques include training **global surrogate models** to mimic the outputs of a black box model. Any conclusions one wishes to draw about the inner workings of the model are instead drawn for the interpretable surrogate. The idea of a global surrogate is quite intuitive, but global surrogates can also fit the model extremely well in certain regimes but poorly in others. In this way, for many explanation methods that only provide a global view of the model, the accuracy of the interpretations may vary unexpectedly across different subsets of the dataset and lose valuable nuance about the underlying model.

- **Local Explanation Techniques.** The four methods discussed above address the challenge of global interpretability. However, these tools cannot be used to characterise why a specific output was produced for a specific data point. Local surrogate models attempt to provide such explanations by building an interpretable model that mimics a black boxed model only in the neighbourhood of a point of interest.

- **Local Interpretable Model Agnostic Explanations (LIME)** approaches this problem by building a linear model in the neighbourhood of a point of interest.[26] To define this neighbourhood, points can either be perturbed in a domain appropriate way (changing individual values by a small amount in tabular data, or changing pixel values in connected components of pictures for image data), or by reweighting other available data points based on a kernel function, which determines how "close" two data points are. LIME and other local surrogate models are intrinsically myopic – they may capture model behaviour well, but only in the region of one specific instance. In addition, while LIME may be able to replicate the output of a model in this region, it does not distinguish between features that cause this behaviour and those that are correlated with the outcome. Other similar, but less popular, local surrogate techniques include **BreakDown**[27] and **High Precision Anchors**,[28] which also create simpler models to approximate local behaviour.

- Shapley values, on the other hand, use game theory to approach model interpretability.[29] In the framework of Shapley values, each feature is assumed to be a player who makes some contribution towards the ultimate model output value. Each player participates in a game by providing a true feature value, and a player who does not participate instead provides some default prior feature value.

  To quantify the importance of a single player, all possible coalitions, or subsets of other features, are created. The method then measures how much (averaged over all coalitions) the prediction made by the model changes when the entire coalition plus the player plays, versus when the entire coalition (not including the player) plays. Interpreting each feature of a model as a player, the method effectively computes how much the prediction changes if that feature is added into the model, averaged over what other features have already been added into the model.

  The core benefit of this setup is that Shapley values adopt an axiomatic view of explanations, where they must assign zero attribution to unimportant features, give equal weight to equally important features, and enforce that features that drive model predictions are always assigned a higher attribution. Shapley values also ensure that the difference between the prediction explained and the average prediction across a set of coalitions is fairly distributed among features, and are the only formulation that satisfies these principles. However, they are also expensive to compute as there are exponentially many coalitions to enumerate over. Popular approximation techniques include **Quantitative Input Influence (QII)**[30] and **Shapley Additive Explanations (SHAP),**[31] which use different sampling techniques to efficiently estimate Shapley values.

  While Shapley values are ultimately calculating the local influence of a set of features to a model score or other desired output, another advantage of these methods is that they can be aggregated from local to global in an intuitive way. Returning to the initial game theory formulation, aggregated "games'' can be created, in which the total "payout" is the average of the model outputs over the entire class. As the assignment of Shapley values is linear, the feature importances for each instance in the class can be averaged to obtain aggregate Shapley values. The latter describes

how each feature affects the prediction on average for a certain class. Extending this class to include the entire dataset thus provides a global interpretation of feature importances.

Finally, Shapley values also allow for contrastive explanations in a way that local and global surrogate models cannot; instead of comparing a prediction to the average prediction across a dataset, one might instead compare it to a subset of predictions or to a single datapoint. This allows for a broader framework of explanation queries (e.g., "why was Jane denied a loan compared to all approved candidates?" vs. "why was Jane denied a loan compared to all men?" or even "why was Jane denied a loan compared to her husband?" should and can yield vastly different explanations).

2.4.2 Post Hoc Interpretability: Model Specific Methods

Model specific explanation methods are particular to specific model classes. Early examples of such techniques are examining coefficients in linear models or generating global feature importances in decision trees by examining Gini impurities of feature splits within a tree.[32]

Nowadays, the vast majority of model specific explanations are concentrated on deep neural networks (DNNs) in tabular, text, and particularly image domains. These methods are typically gradient based. DNN explanation methods that construct purely local views of model behaviour include SmoothGrad saliency maps,[33] guided backpropagation,[34] layerwise relevance propagation,[35] and Grad-CAM.[36] Yet another popular technique is Integrated Gradients,[37] which provides a Shapley value-esque attribution that is both locally and globally faithful by calculating average feature importances with respect to a baseline instance.

Still other DNN explanation techniques focus not only on attributing a model's predictions to input features (e.g., pixels) but also to internal neurons or layers of the network, as is the case with TCAV,[38] or Influence Directed Explanations,[39] which provides a generalisation of many gradient based attributions for arbitrary layers of a DNN. New techniques are arising every day for particular applications; many practitioners conceptualise the inner workings of NLP transformer models by visualising self-attention weights,[40] but there is strong debate in the literature about whether attention maps are reasonable or reliable explanations.[41,42]

2.4.3 Explanations for Inherently Interpretable Models

Models that are "inherently interpretable" limit their architecture in order to be more explainable. These explanations are often argued to be the "most faithful" explanation method as they are "baked" into the model itself.[43] The prototypical inherently interpretable model is the simple linear or logistic regression; with these, feature importances can be intuited from the coefficients of the linear model and the exact way a model reaches its conclusions is clear because a linear model creates its predictions as a weighted sum of input features.

This simplicity can be both a strength and a weakness. Linear models implicitly enforce monotonicity constraints on the model and limit how features can interact with each other. To combat this, generalisations of a linear model that allow for more complex feature interactions exist. Examples include the generalised linear model (GLM), which generalises how an output depends on a feature value to non-Gaussian distributions;[44] the generalised additive model (GAM),[45] which relaxes the

condition that only linear relationships between inputs and outputs can exist; and methods like GA2M[46] and RuleFit,[47] which allow models to capture low-dimensional feature interactions.

Decision trees, decision sets,[48] rule based classifiers[49,50,51] and scorecards are also considered inherently interpretable models in that they force models to create predictions via clearly defined rules. More recently, there has also been some research to add flavours of interpretability constraints directly into model training itself (e.g., Bayesian rule based reasoning models,[52] sparsifying neural networks through gradient penalties,[53] or even by adding human feedback into an optimisation procedure[54]).

2.4.4 Counterfactual Explanations

So far, this section has focused on various explanation methods that attempt to understand model behaviour with respect to a model's constituent features. These techniques also have provided reasoning: an explanation of why a model behaved a certain way, and not necessarily how to change this behaviour. As described in Section 2, data subjects often want actionable guidance on redress opportunities. They seek explanations via a set of example actions they could take to improve their outcome – for instance, by finding data points in the vicinity of a given instance that have different outcomes.

These types of explanations are called counterfactuals. Counterfactual explanations essentially "interrogate" a model to probe for required changes to input features that would result in different model decisions. These explanations can be used by model developers to sanity-check the trustworthiness and fairness of a model, or even presented to end-users/customers in order to justify model behaviour. Each counterfactual explanation consists of a sequence of changes the data subject might make to improve their outcome, e.g., by closing a credit card or reducing the amount requested in order to secure a loan. Desirable attributes[55] of a good counterfactual explanation are as follows:

- **Validity.** The counterfactual must be assigned the desired class by the model (e.g., "reducing the loan amount requested by $10,000 will certainly cause the loan to be approved").
- **Actionability.** Any proposed changes must be feasible for an individual (e.g., individuals cannot suggest "reducing your age will improve the chances of you getting a loan").
- **Sparsity.** Proposing a change to a large number of features can result in actionable guidance which is hard for a human to interpret.[56] Counterfactual explanations should therefore mutate as few features as possible in order to be more understandable and actionable.
- **Closeness to the data distribution.** Even if a proposed change is actionable, the new data point that would result from making this change may lie outside the training data's distribution. As a result, the model's predictions on the new data point would be unreliable (e.g., if a model is trained on people with incomes between $50,000 and $100,000, we should not suggest that a user increase their income to $150,000).
- **Causality.** Counterfactual suggestions should account for correlations between features that would have to change in tandem (a proposal that an individual should change their desired loan repayment period to five years from 10 years should lead to doubling their desired monthly repayment or halving the loan amount).

- **Privacy.** Giving a user a set of proposed changes should not reveal other people's underlying data (we cannot suggest that a user simply change themselves to look like someone else with a different outcome) or violate privacy laws. Privacy is often overlooked in the counterfactual explanations literature because it poses technical challenges[57] by inhibiting how much information can be "given away" to the user.
- **User preference solicitation.** A good counterfactual explanation would incorporate user preference in some way (e.g., suggesting "close three credit cards" might be much easier for a customer than "increase your income by $200"). These preferences are difficult both to capture and express, but are critical to users taking advantage of redress mechanisms.

Numerous techniques exist to generate counterfactual explanations,[58,59,60,61] but few meets all of the desirable attributes. Open source frameworks like DICE[62] and FACE[63] exist, but fail to satisfy privacy, for example. There is ongoing research in the field to create counterfactual explanations that are of suitable quality.

2.5 ML Explanation Method at Work: Illustrative Example

Many FSIs have started experimenting with one or more of the methods outlined above in building internal transparency. This subsection illustrates how one popular technique – using Shapley values to calculate feature importance – can be used in the context of a banking credit example.

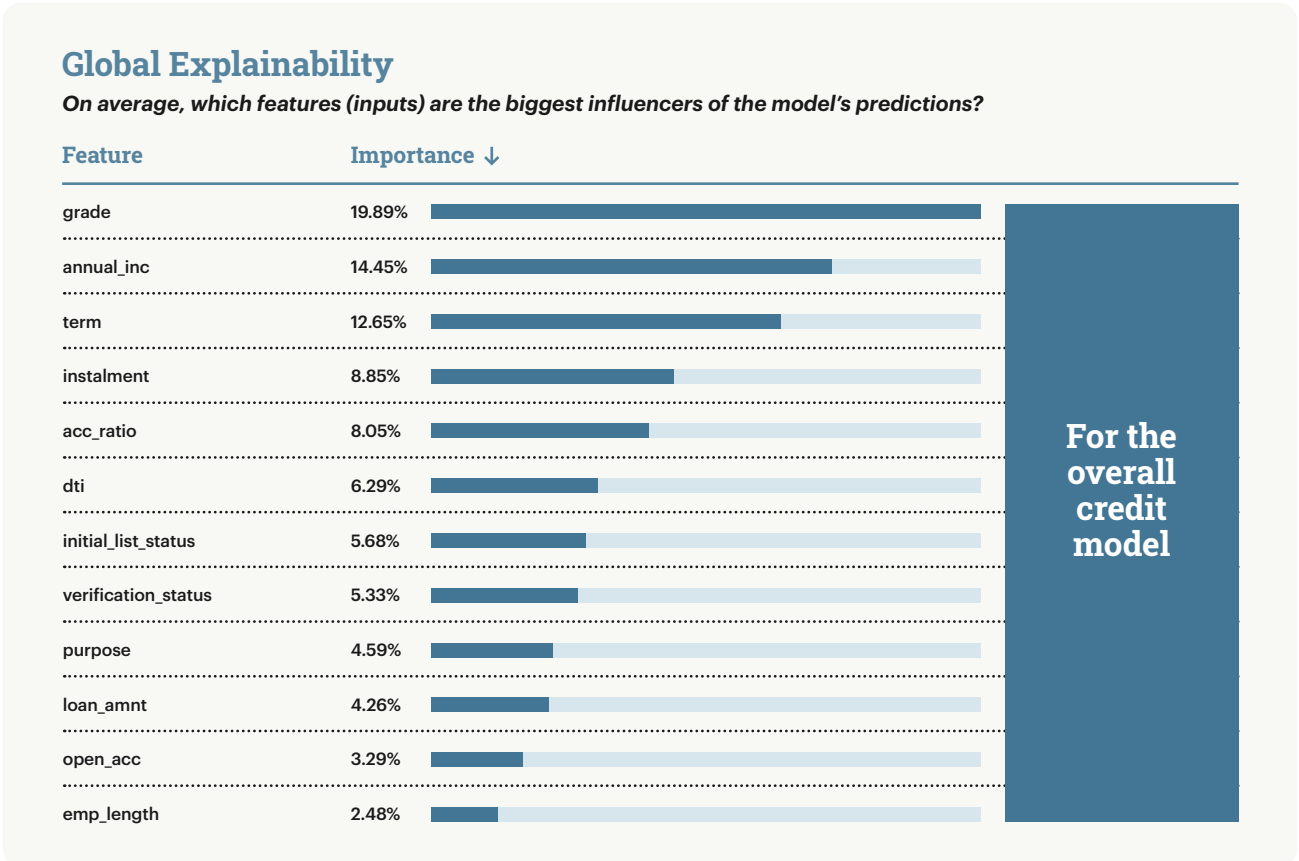


Figure 2.1: Feature importance (Global)

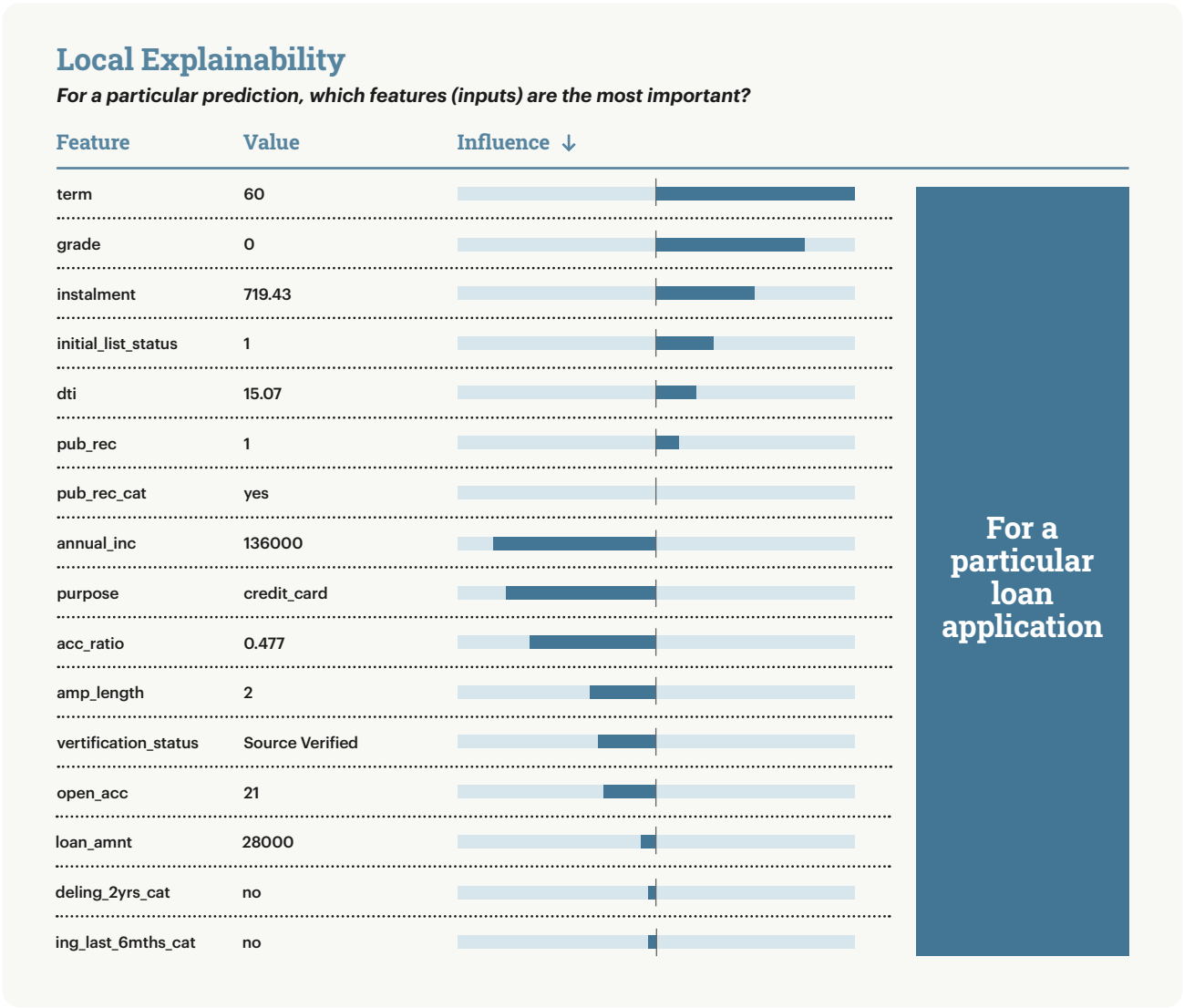


Figure 2.2: Feature importance (Local)

Beyond a simple assessment of the relative importance of different features, Shapley value importances can also be plotted against the value of a feature to provide additional insights into the conceptual soundness of the model. In the example below, a credit expert is able to review how changes in the value of the input variable 'dti' (debt to income ratio) results in changes to its marginal influence on the model outcome. The area highlighted in the bottom righthand side of the graph represents unexpected behaviour suitable for further investigation: a high dti ratio causing an improvement in the applicant's perceived risk.

Conceptual Soundness

Influence Sensitivity Plots

- Influence sensitivity plots show how the model uses a feature as a function of its value.
- Can be used to verify intuition regarding influence monotonicity, smoothness, islands of atypical influences and more.
- Goes beyond traditional analysis for linear models to capture non-linear relationships and feature interactions within sensitivity plots.

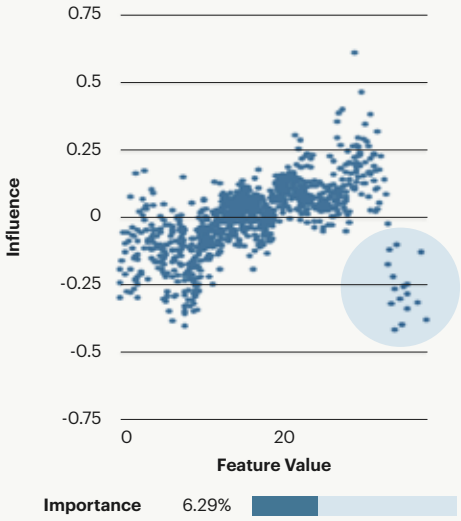


Figure 2.3: Conceptual soundness (1/2)

1	**Is the feature influence behaviour monotonic?**	Features such as income and savings account balance are expected to show a monotonic behaviour.
2	**Are there unexpected jumps in the feature influence plot?**	Jumps in the trend indicate that models are internally bucketing a feature.
3	**Are there plateaus or is there step behaviour?**	Most features exhibit a plateau at extreme values due to diminishing returns of risk beyond a certain point.
4	**Is the behaviour of this feature reasonable to a subject matter expert or is it unexpected?**	If behaviour is unexpected, there may be a data or model problem.

Figure 2.4 Conceptual soundness (2/2)

03 Methodology

The previous section outlined the key concepts around external and internal transparency. This section aims to provide a "how to" guide to FSIs on operationalising those concepts.

The proposed methodology takes the form of a series of "checklist" questions mapped to the steps in a typical AIDA lifecycle, outlined below in Figure 3.1.

- The original FEAT document outlines the Principles to be used to guide FSIs' work on Transparency (Principles 12, 13, 14).
- FSIs should first translate these Principles into appropriate internal standards – not specific to individual use cases but applicable to the FSI as a whole.
- Teams accountable and responsible for individual AIDA systems should then adopt and operationalise the appropriate standards during development, validation, deployment and ongoing monitoring of their AIDA systems.

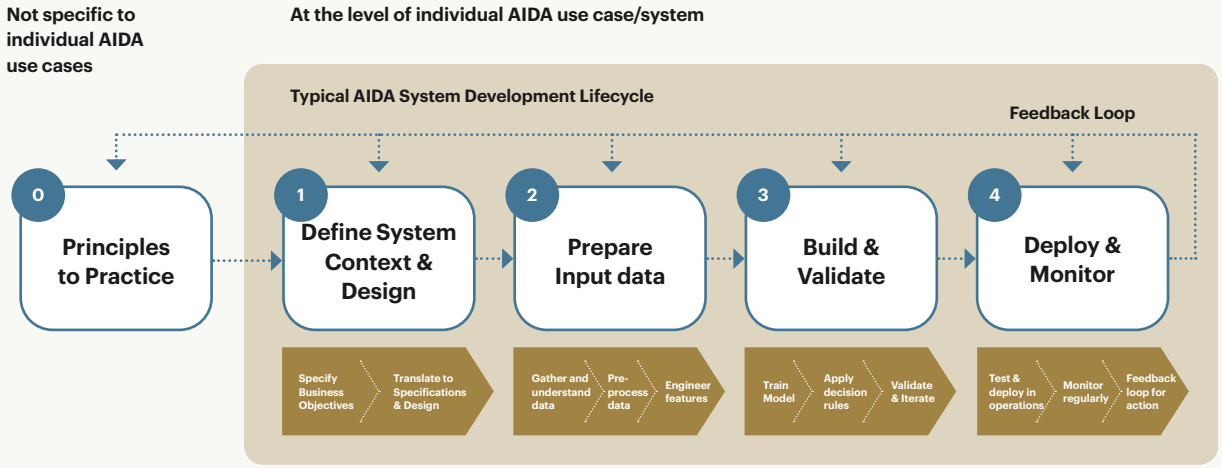


Figure 3.1 AIDA system development lifecycle

Note that for both external and internal transparency, explanations should cover the final outcome of the AIDA system after any planned human interventions have been made. In addition, for internal transparency, stakeholders may also be interested in explanations of individual ML models that form part of the AIDA system, prior to planned human interventions.

3.1 Step 0: Set (Transparency) Standards Internally Within the FSI

Step 0 is not specific to an individual AIDA use case or system. While the appropriate level of transparency may vary by use case, an FSI would ideally have in place a common set of internal standards upfront to help guide such decisions on the appropriate levels of transparency (and associated accountability) in Steps 1-4. Otherwise, each team embarking on an AIDA project will need to come up with its own decision framework. Defining such standards can provide a level of consistency across all use cases in the FSI (including the potential for common tools if appropriate), while still allowing for tailoring based on materiality. Such standards should be guided by prevailing legal and regulatory requirements and by the FSI's own data/AIDA ethics principles.

The five transparency 'standard' questions to be answered in this step are as follows:

- (T1) Has the FSI defined the factors it will use to determine whether external transparency is essential for a particular AIDA use case?
- (T2) (Where an FSI has chosen to provide external transparency) At each stage of the FSI's customer lifecycle, has the FSI determined what proactive or reactive communication may be needed, and the standard templates/interfaces for the same?
- (T3) Has the FSI defined the factors it will use to determine the extent of, and audience for, internal transparency for individual AIDA use cases?
- (T4) Has the FSI defined an acceptable set of AIDA ML explanation method(s) for use within the firm?*
- (T5) Has the FSI set minimum accuracy standards for such explanation methods?*

These questions should be answered once when setting up the initial standards for the FSI's overall use of AIDA systems (i.e., they not specific to a use case), and then reviewed periodically (e.g., once a year) to ensure ongoing alignment with evolving technology (e.g., changes in "state of the art" when it comes to explanation methods), regulatory thinking (e.g., new standards on minimum information to be shared with customers) and internal/customer feedback (e.g., results of Net Promoter Score surveys or other forms of primary feedback from customers on the explanations provided to customers).

With regard to T1-T2, it is unlikely that an FSI will choose to apply different standards for AIDA driven decisions compared to other traditional decision making processes. For example, whether a credit decision is based on a traditional scorecard or a new ML model, the FSI will probably have similar approaches to determining the level, timing and form of the customer facing explanation.

May vary by use case type and/or materiality. At the time of writing this report, industry thinking around explainability standards is still evolving, so it is likely that many FSIs will not be in a position to meet this requirement straight away. Additional work is likely to be needed in this space in the coming months.

3.1.1 Factors to Determine the Need for Explanations to Data Subjects

This section sets out considerations to help FSIs assess the need for reactive explanations to data subjects in the context of FEAT Principles 13 and 14 (i.e., in response to a request for information on a specific AIDA driven decision). The starting premise of this methodology is that – all other things being equal – such requests should be fulfilled by the FSI.

In practice, one set of factors makes such "full transparency" inappropriate and/or impractical, while others increase the importance of such transparency. An FSI will need to arrive at a final decision on whether to share information with data subjects in a particular use case by balancing these two sets of considerations. This section provides FSIs with a checklist of such considerations, so that they:

1. Can make the right decision in each AIDA use case.
2. Are able to justify that decision to internal stakeholders (e.g., risk, compliance, etc.) and to regulators if needed.

Like with other aspects of the Veritas methodology, these considerations are not meant to replace regulatory requirements around customer or regulatory disclosure.

Wherever there is an explicit regulatory requirement in a jurisdiction to ensure transparency to data subjects for a particular use case, that will take precedence over such considerations.

A. Arguments for Transparency in AIDA Driven Decisions

Prospects or customers impacted by the use of AIDA should be provided information about the data and logic used to make such decisions if it has the potential to:

1. Prevent the data subject from accessing a financial product or service altogether (e.g., their ability to get credit or insurance cover, their ability to access certain investment products or even their ability to enter into a relationship with the FSI). However, transparency requirements may not apply when such denial is due to concerns about the financial crime or fraud risk posed by the data subject.
2. Prevent the data subject from enjoying one or more benefits that they can reasonably expect from an existing product or relationship (e.g., their claims against an insurance policy, their ability to make payments or other transactions). It is worth noting that transparency requirements may not apply when a denial is due to concerns about financial crime or the fraud risk posed by a data subject, but should continue to apply when the denial is due to fraud or security concerns not originating from the data subject (e.g., "we stopped you from making this payment because we believe you were going to be subject to an online scam").
3. Risk breaching the FSI's duty of care to them (e.g., their right to be not sold a product or service that is unsuitable for them, their right to appropriate treatment in case of specific vulnerabilities such as when being pursued for collections or repossession, and their right to have complaints reviewed and addressed as per agreed terms).

4. Risk breaching the data subject's personal data rights (e.g., incremental disclosure of protected data or inappropriate engagement with customer that go against previously agreed terms and conditions , such as when a data subject's video call with the FSI is being used by an algorithm to detect emotions, and explicit consent has not been sought for the same).
5. Expose the data subject to unfavourable commercial outcomes compared to others when applying for or using a product or service (e.g., pricing, penalties, product conditions such as tenure or level of collateral).

An adverse impact of the type outlined above can come about directly (due to the use of AIDA in the context of a specific use case) or indirectly (where the outcome from an AIDA system is used beyond its original context). An example of the latter would be the use of an AIDA driven credit score calculated during a loan application in a subsequent insurance underwriting application.

Note that conditions one through to four apply even in absolute terms, but become particularly important in situations where the data subject is disadvantaged compared to other data subjects, or groups thereof. On the other hand, condition five only applies in the context of potential discrimination suffered by the data subject rather than as an absolute right.

While the FEAT transparency requirements are typically understood to relate to an FSI's relationship with its customers, FSIs also apply similar considerations in situations in which their use of AIDA systems can impact other people, such as staff and contractors. For example, impacted staff (or prospective staff) could seek transparency when AIDA driven decisions impact their ability to be hired, evaluated, promoted or rewarded at work, or have their employment terminated (except where the confidentiality of such decisions is protected, even for human decision making, due to other legal or regulatory requirements such as employment or "whistle blower" protection laws).

The arguments highlighted above are defensive – i.e., as a means of avoiding harm to prospects, customers or other data subjects. However, FSIs should also consider instances in which transparency might be used to improve the customer experience. For example, instead of just explaining why a prospective customer cannot get a particular level of insurance cover, a well-designed communication of the reasons for the decision, combined with an alternative preapproved offer, may end up increasing trust and customer satisfaction.





B. Arguments Against Transparency to Data Subjects in AIDA Driven Decisions

Ranged against the positive indicators for greater transparency in AIDA decision making is a series of contraindicators that could constitute valid arguments for not sharing information with data subjects. These include:

1. **Legal restrictions:** where providing information as requested by the data subject would require the FSI to share its suspicions on the data subject of fraud, sanctions breach or money laundering; or require it to disclose the personal data rights of another data subject who may have received a more favourable outcome. However, the latter will not apply if the relevant information on the other data subject is already in the public domain, or has been voluntarily disclosed by them.
2. **Concerns over "gaming":** where there is a risk of unscrupulous individuals being able to use one or more transparency requests to work out how to bypass the FSI's mechanisms to manage customer or transaction specific risk (e.g., detecting/preventing fraud, cybersecurity, credit/underwriting or reputational risks).
3. **Competitive concerns:** where there is a risk of the information being disclosed through such transparency requests adversely affecting the company's intellectual property (e.g., a proprietary insurance underwriting algorithm that allows a firm to underwrite risks other insurers cannot) or financial stability (e.g., an internal freeze on approving new loans due to broader credit, liquidity or other risks unrelated to the data subject's specific application).

FSIs will need to meet a high bar in making arguments two and three. A one-off request by a single impacted data subject on the reason for denial of credit is unlikely to qualify as an attempt to game the system, although multiple requests by the same individual or a group of individuals at the same time might qualify.

3.1.2 Recommended Proactive and Reactive Communication at Each Stage of the Customer Lifecycle

Each FSI is likely to have its own definition of the customer lifecycle; here is an illustrative example:

1. **Prospect engagement:**
   engaging a potential customer as part of a marketing or sales campaign.
2. **Cross-sell/up-sell/retention:**
   engaging an existing customer to retain them or to sell them a new product or solution.
3. **Application process for product/solution:**
   engaging a prospect or customer when they want to buy a particular product or solution, or to change terms for an existing one.
4. **Approval or rejection of application:**
   engaging a prospect or customer to convey the FSI's decision with respect to their application, and to address any complaints/redress requests from them.
5. **Fulfilment of existing product/solution:**
   providing the solutions expected as part of an existing relationship could involve explicit decisions such as those around acceptance (or rejection) of insurance claims.
6. **Ongoing customer service in existing relationships:**
   addressing customer queries/requests related to their existing relationships/products, and handling complaints where needed.
7. **"Hardships":**
   managing aspects such as collection/recovery and asset repossession when a customer is unable to meet their obligations to the FSI.
8. **Exit:**
   termination of customer relationship or individual product/solution.

As part of the transparency standard setting exercise, the FSI should determine the appropriate level of customer facing explanations at each stage of the customer lifecycle.

Stage	Proactive transparency if AIDA is used*	Reactive transparency if AIDA is used**
1 **Prospect engagement**	• The fact that AIDA is used in the decision to reach out to data subject. • Data subject's right to be excluded from such communication.	-
2 **Cross-sell/ up-sell/ retention**		-
3 **Application for product/solution**	• The fact that AIDA is used in decision making. • Human oversight on AIDA use • Types of data used.	-
4 **Approval/ rejection of application**	• The fact that AIDA is used in fulfilment (e.g., insurance claim). • Data subject's rights around transparency/ redress.	• Reasons and/or actions in case of adverse AIDA driven decisions. • Data used by the AIDA system. • Impact of AIDA driven decision on data subject.
5 **Fulfilment of existing product/ solution**		
6 **Ongoing customer service**	• The fact that AIDA is used in providing services (e.g., chatbots, image based authentication).	• Reasons for a specific AIDA driven communication or action about which the data subject is unhappy. • Data used by the AIDA system.
7 **"Hardships"**	-	
8 **Exit**	-	

Table 3.1: Proactive and reactive transparency across the customer lifecycle

Section 2.2 highlighted the need to ensure that communication content aimed at data subjects is meaningful for the receiver. Relevant stakeholders within an FSI (e.g., customer experience, risk, compliance, legal, etc.) may want to set internal standards around how much of it should be shared with the data subject in one go (vs. provided incrementally), and in what exact form. For example, one FSI prefers starting with a visual summary of reasons and an interactive user interface which allows data subjects to visualise the impact of changes on the final outcome. Another shuns "actions" altogether and instead offers an alternative product or service that is preapproved for the data subject.

** Traditionally through customer/product level Terms & Conditions, but increasingly through more intuitive disclosures during customer journeys.*
*** Could be in real time (e.g., in an app or online portal) or in response to complaints through email/contact centre/branch.*

Here is an example, shared by Microsoft and HSBC at a US Consumer Financial Protection Bureau tech sprint event, of a single dashboard that incorporates both reasons and actions.[64]

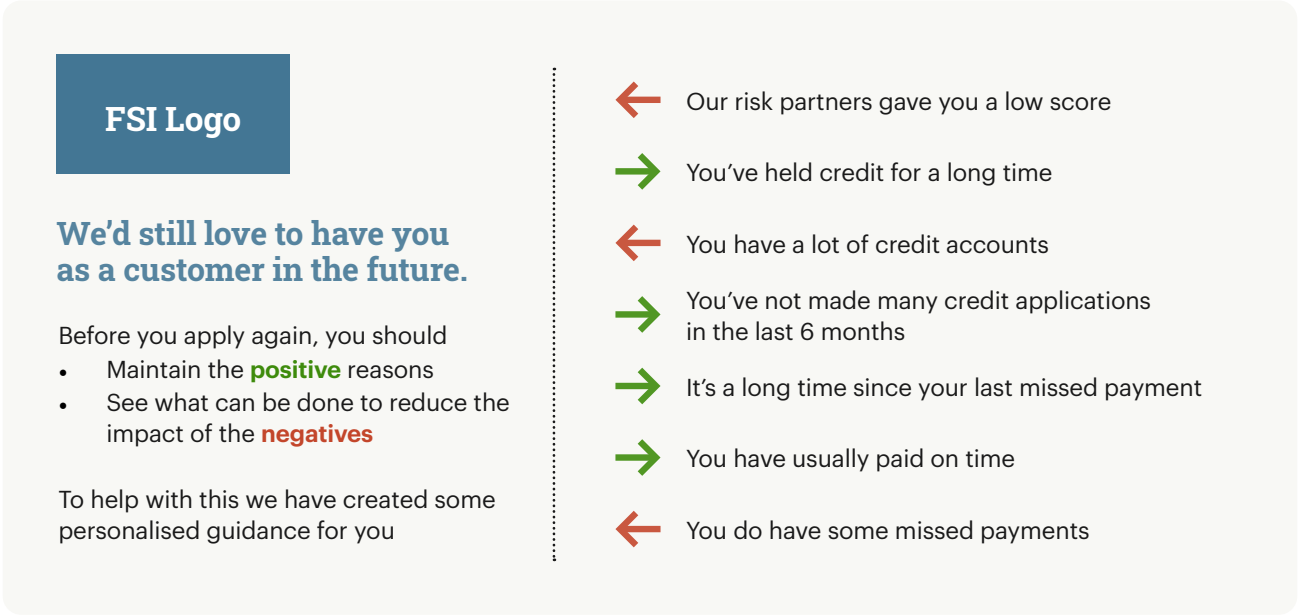


Figure 3.2: Sample format: customer facing explanations (1/2)

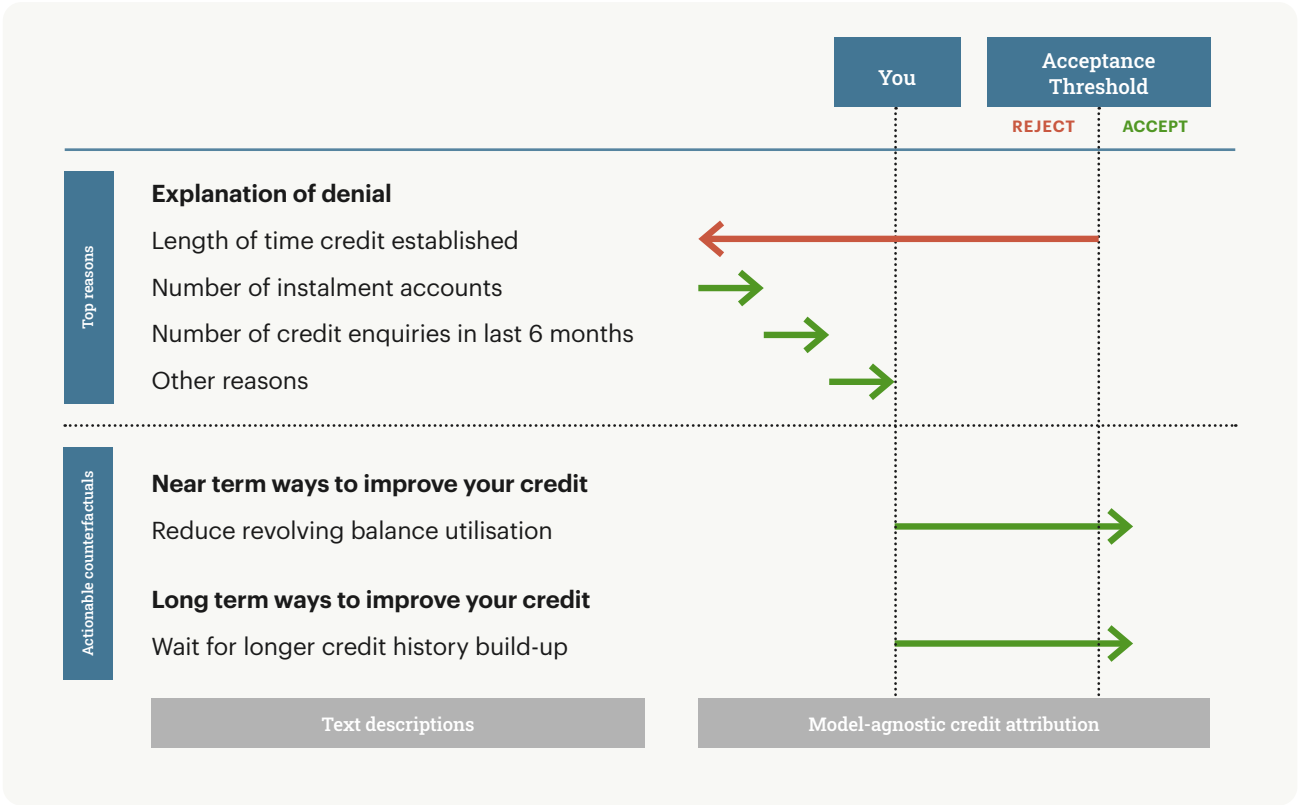


Figure 3.3: Sample format: customer facing explanations (2/2)

3.1.3 Factors to Determine Internal Transparency Requirements

Even where an FSI uses the above methodology and decides to provide partial or no information in response to a data subject request, it needs to still make sure that the necessary information is available to internal stakeholders within the FSI. This availability will enable reviews by internal risk, compliance and audit teams if required to provide assurance on the justifiability, fairness and ethics of the AIDA driven decisions. It will also provide the necessary auditability for subsequent regulatory or legal investigations, if any.

The extent to which such transparency is needed, and the audience for the same, will be determined by the materiality of the AIDA use case. Different audiences – e.g., data science team managers, business stakeholders, risk and compliance teams – will require different levels of technical detail from the explanations.

3.1.4 Acceptable ML Explanation Method(s) for Use Within the FSI

Section 2 highlighted the various explanation methods available today for ML AIDA systems. To ensure consistency, an FSI might choose to define internal standards listing our acceptable explanation methods for ML AIDA systems, perhaps allowing for differences by materiality. In doing so, it should keep in mind the following desirable attributes of a "good" explanation methodology:[65]

- **Fidelity.** Measures how well the explanation approximates the prediction of the model, or its "accuracy" in a sense. Some techniques like local surrogate models only offer local fidelity, as they can only elucidate a model's behaviour in a narrowly defined region. High fidelity explanations, on the other hand, are able to extrapolate further and separate correlation from causation.
- **Consistency.** Measures the similarity of explanations across models trained on the same task, or even a single model explaining two different scopes (i.e., local vs. global). Explanations cannot be compared against one another if inconsistent, which is why explanation techniques that establish consistency in their local and global views (e.g., Shapley value based attributions) can be more broadly applied.
- **Comprehensibility.** Explanations are useless unless they are meaningful to the humans who are using them. Unless humans can work with and reason about an explanation, it is not aiding in understanding a model's behaviour.
- **Accuracy and certainty.** Many explanation strategies are approximations of fundamental quantities, or may not capture a model's confidence in making the decision. Explanation methods in practice should be as accurate as possible. One way of measuring accuracy could be to require guarantees of the form "X% of exact explanations Y% of the time".
- **Flexibility.** Explanation strategies should be generalisable enough to provide comparative insights across a variety of population groups. For example, we should be able to explain why a decision was made for a certain person, but also why that decision may have been different for other similar individuals.

Explanation methods that are derived from these principles of a "good" explanation can provide more consistent, accurate, and flexible explanations than lower fidelity methods that focus on local regions. Inherently interpretable architectures satisfy many of these attributes, but come with the consequence of intrinsically limiting a model's architecture and ability to learn complex feature interactions. Among post hoc explanation methods, a model agnostic one may allow for an easier comparison of explanations between different model classes (i.e., when comparing an incumbent ML model to a traditional one that has been in use). However, model specific methods may be attractive in the context of certain types of deep neural networks.

3.1.5 Minimum Acceptable Accuracy Standards for ML Explanation Method(s) Within the FSI

Explanation outputs can dramatically and meaningfully differ based on the comparison group and the output type that is being explained. FSIs – particularly those with extensive adoption of AI/ML – may want to consider introducing internal standards around the accuracy and consistency of explanation methods.

3.2 Step 1: Define System Context and Design

This is the first step in the lifecycle of an individual AIDA use case, represents an opportunity for the AIDA system owner to apply the FSI's internal standards, and determines the appropriate level of transparency needed for the specific AIDA use case.

At this stage, the team leading the design of the AIDA use case should have adequate information on its purpose, target operating model, and relevant external/internal audiences to answer the following questions for the specific use case:

- (T6) Has the AIDA use case team determined whether there is a need for external (customer facing) transparency? *Apply standards from T1 to help answer the question.*
- (T7) If yes, has the team identified the proactive and reactive communication needed at each stage of the customer lifecycle, and the form of such customer facing communication? *Apply standards from T2 to help answer the question.*
- (T8) Has the team determined the level of internal transparency needed, and the audiences for the same? *Apply standards from T3 to help answer the question.*
- (T9) Has the team selected a suitable explanation method for this specific use case from the approved list in T4?
- (T10) Has the team ascertained that the chosen explanation method/implementation meets the minimum accuracy requirement for this specific use case (based on T5)?

In each case, the team leading the use case should record not just their decision, but also the justification for the same. Note that the answers and justifications may get refined in Step 3 once more information on the actual implementation becomes available.

3.3 Step 2: Prepare Input Data

This is the second step in the lifecycle of an individual AIDA use case. No transparency related questions are proposed for this step.

However, the team leading the specific AIDA use case should make sure that relevant information about the training data is recorded (e.g., source, vintage, consent from data subjects, dictionary and known limitations of the data such as those arising from data quality and representativeness tests). This information will be relevant across the various FEAT Principles.

3.4 Step 3: Build and Validate the AIDA System

This is the third step in the lifecycle of an individual AIDA use case. By this stage, details of the actual implementation of the AIDA system will be clear. The team leading the use case should take this opportunity to revisit its original answers to T6-T10. For example, it may choose to modify the choice of ML explanation method if a more complex ML algorithm (compared to the original design) is being used. Similarly, if a decision has been taken to increase the level of "human in the loop" involvement, it may be possible to relax the levels of minimum accuracy for explanations originally agreed.

Once this is done, the focus should move to the operationalisation of the transparency requirements. The following questions can act as a checklist:

Internal Transparency

- (T11) Have internal transparency dashboards/reports been implemented in line with the requirements agreed in T8-T10?
- (T12) Have relevant first and second line control teams – including model validation where relevant – reviewed and approved these outputs (e.g., local and global explanations, conceptual soundness, etc.)?
- (T13) Where the explanations have not met first/second line expectations, have appropriate mitigation actions been taken? (e.g., switching to a simpler model despite a reduction in predictive accuracy, dropping difficult to explain features, introducing more human oversight, etc.)

External Transparency

- (T14) In line with the external transparency requirements agreed in T6-T7, has appropriate system functionality been developed and tested as part of the AIDA system's implementation plan?
- (T15) Have operational processes such as customer service and complaint handling been modified appropriately to incorporate AIDA customer transparency? Have relevant staff been provided with appropriate training to address customer queries?
- (T16) Have customer/website Terms & Conditions been appropriately updated (i.e., to explain how data is used, how benefits and risks to individuals are associated with the processing, and how individuals may participate and object where appropriate)?

Common to Both

- (T17) Does the AIDA system implementation support the agreed internal and external explanations even after "go live" (i.e., not just as a one-off before approval but throughout the lifetime of the AIDA system)

3.5 Step 4: Monitor AIDA System

The actual assessment of AIDA system monitoring requirements should have been completed in Step 3. In this step, the focus should be on ongoing monitoring.

04 Beyond Individual Use Cases: Achieving AIDA Transparency at Scale

The last two sections contain a rich conceptual basis as well as extensive practical illustrations to enable FSIs to ensure appropriate levels of transparency in an AIDA use case. However, AIDA usage can only be scaled up if such methods become part of day-to-day processes and systems at FSIs. This section makes five recommendations in this regard:

1. **Embed AIDA transparency considerations into existing risk frameworks.** Such as those around model risk (or its equivalent outside banking), data privacy and the fair treatment of customers.
2. **Set out your stall.** FSIs should provide a proactive, consistent message on its approach to the use of data and algorithms, and its impact on customers and staff. This could be through websites and other electronic and physical channels.
3. **Overinvest in training.** Of customers, staff, management and risk/control functions. The more AIDA can be demystified, the easier it is to make it trustworthy.
4. **Supplement use case level investments in tools for AIDA transparency with more systematic support across use cases.** For example, a single customer level data dashboard that provides an overview of all data held by the FSI about a data subject, and where it is used across the FSI; standardised tools for AIDA explanations that can be used across use cases; a common AIDA model repository that that includes basic information about each AIDA system in one place.
5. **Don't view transparency as a standalone topic.** While important in its own right, transparency can have a much broader impact on the trustworthiness of AIDA. Done right, it can underpin other aspects of AIDA quality, such as fairness and stability. It should be embedded throughout the model lifecycle, and not be seen as a "one-off" task towards the end of the model development, testing and/or validation steps.

The recommendations included above are aligned with section 3.2.1 of the Veritas Document 3 FEAT Principles Assessment Methodology document and detailed explanations are provided from transparency perspective.

4.1 Make the Best Use of Existing Risk Management Frameworks

While the use of AIDA may be growing, FSIs already have the basic frameworks to meet their obligations around AIDA driven decision making.

- Model risk policies and standards, or their equivalents in insurance and asset management, provide a useful vehicle to embed internal transparency standards.
- Policies and standards around fair treatment of customers, and associated concepts such as the FSI's duty of care to customers, provide the necessary vehicle to embed considerations around customer facing transparency.
- Data Privacy policies and standards in most FSIs already require a level of transparency for data subjects around the use of data and algorithms, and the availability of recourse options.

Creating a new, dedicated framework for AIDA transparency (or for FEAT more broadly) in isolation from these existing policies and standards is not advisable. Wherever possible, FSIs should focus on enhancing the policies, standards and processes underpinning existing risk frameworks to meet the requirements around AIDA transparency.

4.2 Set Out Your Stall

Some FSIs have begun publishing "charters" around data ethics, which proactively articulate key principles and guarantees around their use of data and algorithms with customers (and other stakeholders). Industry associations such as the Institute of International Finance (IIF) have also proposed similar charters at a sectoral level. Such initiatives can help FSIs improve the efficiency and effectiveness of the proactive component of their AIDA transparency obligations. They are also being used by some firms, particularly in the technology sector, to attract talent. Additionally, such charters can also help provide internal guidelines to teams across the FSI as they scale AIDA usage.

4.3 Overinvest in Training

FSIs should train frontline staff or representatives around the use of AIDA in engaging with and making decisions relating to customers. Often, it is branch/contact centre staff, (insurance) agents or financial advisors who will be at the receiving end of AIDA related queries from prospects and customers. They should also have access to systems that provide them with the information they require in a manner that both they and the prospects/customers can understand. Such training, and tooling, should also be provided to the next level of staff (e.g., underwriters, claims assessors, credit approvers) who may be called upon by frontline staff to address such queries.

Training and education should also be extended to business executives who are not AIDA specialists but must take accountability for the outcome of AIDA systems. Ensuring that senior leaders understand how the AIDA system works, and where it may have deficiencies, is essential for them to demonstrate accountability.

4.4 Create Standards and Tools for AIDA Transparency that Extend Across Use Cases

Instead of leaving each AIDA use case owner to come up with their own approach, some FSIs have started investing in standards, and supporting technology capabilities, that allow teams across the organisation to meet their AIDA transparency obligations. Examples include:

A. Customer Level Personal Data Dashboard

Building on a concept implemented by some technology firms, some FSIs have attempted to create a single dashboard that allows all customers to see what types of data the FSI holds about them (e.g., demographic information from initial onboarding, information gleaned from day-to-day transactions or engagement with the FSI, information purchased from third parties, etc.). This dashboard can be linked to major categories of AIDA use cases, so that the customer is able to see an overview of the kinds of data being used to make the different types of decisions that affect them.

Creating a dashboard like this remains difficult for most FSIs, due to the fragmented nature of their technology landscape, and the fact that customer data and consent is rarely centralised in a single system. However, even by making initial moves towards an ambitious goal like this, FSIs may be able to dramatically streamline their proactive communication with customers on data and algorithm use.

B. Central AIDA Model Repository

Many FSIs are already attempting to centralise their AIDA models into a single repository. While this approach has numerous efficiency and risk management benefits for the broader model lifecycle, it has a particular application for AIDA transparency. In addition to keeping an up-to-date inventory of all AIDA models in use, such a repository can provide a place where model owners can document basic metadata about each model. Such metadata could include:

- **(Information on) Training data**
  - Data used: source, vintage, consent.
  - Data not used (considered or available but ultimately not used).
- **Variables**
  - Transformations & dimensionality reductions.
  - Inferences/proxies.
- **Method (e.g., Support vector machine, decision tree, etc.)**
  - Used
  - Not used (tested but not used due to accuracy, reliability, etc.)
- **Scope (decisions made)**
- **Scale (number of decisions:** one per customer or employee, multiple per customer or employee, sometimes on some customers).

C. Standards and Tools for Model Explanations

As articulated in Section 3, the field of AIDA model explainability, particularly for ML models, continues to evolve rapidly. FSIs should seek to centrally specify one or more approved explanation methods, so that there is a minimum level of consistency in the way in which individual data science teams generate internal and external transparency. Many explanation methods can already be designed and implemented to provide accuracy guarantees of the form "X% of exact values Y% of the time". When combined with more detailed feature influence plots, they should provide FSIs the necessary level of confidence in their AI/ ML models.

4.5 Don't View Transparency or Explainability in Isolation

Viewing transparency or explainability in isolation misses the bigger picture.

Transparency is rarely an end in itself. Instead, it can serve as a greater mechanism to test, debug, validate, and monitor models. Transparency into an AIDA model's ways of making predictions is useful, but the broader end goal must be to achieve high model quality and trustworthiness. Model quality in this case is much more than the accuracy of the results. Validating a model's conceptual soundness, probing it for signs of unfair bias, and inspecting its stability on new data are some of the model quality analyses that can be enabled by robust and accurate explanations. These are necessary inputs to engender trust in the AIDA models and the FSI itself.

In addition, the effort to build transparency should be thought of as something that impacts and is involved with every step throughout the model lifecycle. A key driver of the surprisingly long time it takes to deploy AIDA models in production is the disconnect between model development and the subsequent steps of model review/validation and buy in from businesses. Embedding explainability early into the lifecycle of the model can help data scientists continuously debug and improve their models well before formal review or validation. Also, many problems with AIDA models start only after they have been deployed, when shifting data or data relationships start affecting results. To maintain ongoing trust in AIDA models, it is critical to also have the ability to explain and debug AIDA models after they go live.

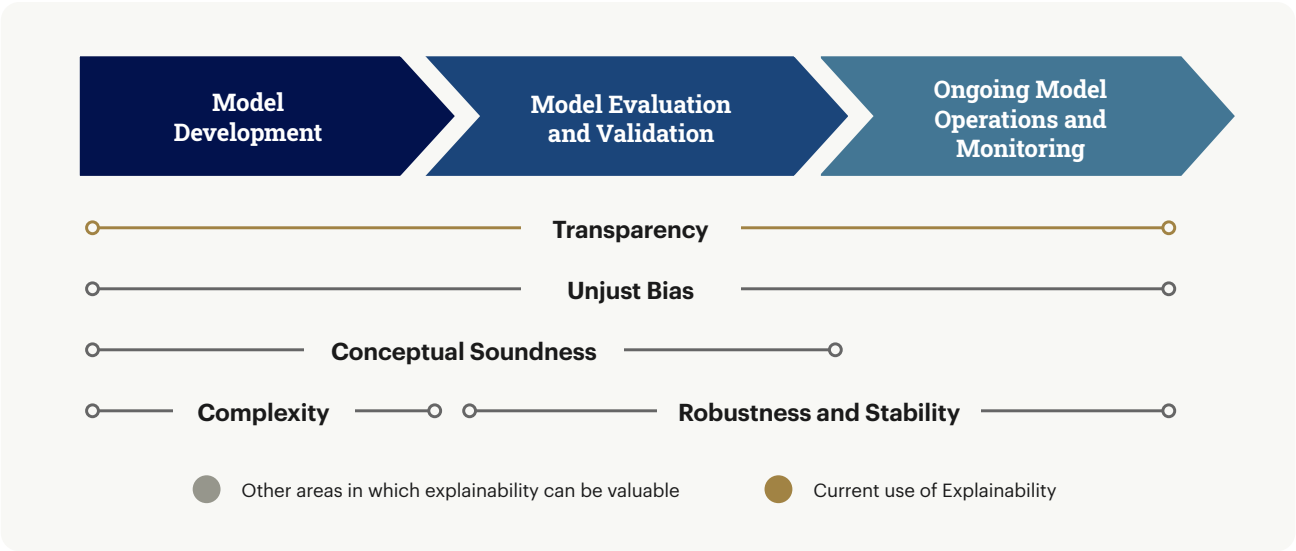


Figure 4.1 Embedding transparency into AIDA system development lifecycle

05 Topics for Future Consideration

In creating this document, the Veritas Transparency Workstream has identified several areas in which greater clarity, through regulatory mandate or industry level initiatives, will be useful.

- **Alignment between AIDA and 'pre-AIDA' transparency requirements.** Many FSIs have pointed out that some aspects of the FEAT guidelines on transparency go significantly beyond the existing (pre AIDA) requirements for customer communications. As highlighted in section 1, it is difficult to isolate the standards and processes for customer facing transparency for AIDA driven decisions from those used for other types of decisions. Therefore, it may be impractical to enhance requirements around customer transparency for AIDA driven decisions without similar enhancements for non-AIDA decisions.
- **Second order impact of AIDA driven decisions.** FEAT Principle 14 requires FSIs to inform data subjects, on request, about the impact of AIDA driven decisions on them. Obviously, FSIs will inform data subjects wherever the AIDA driven decision results in an immediate adverse outcome for them, such as denial of loan or insurance claim. However:
  - Does the FSI have the right to use the results of that AIDA driven assessment in any other use cases (e.g., a future loan application or insurance claim), and/or to share such results with their partners (e.g., with an external credit bureau)?
  - Even if that is the case, should they allow data subjects to know about such "second order" impacts of the AIDA driven decision on them?

  Current thinking is that such calculated results be treated as "derived" data and would be subject to prevailing consent rules around such data.
- **Treatment of AIDA usage via third parties.** AIDA developed or provided by third parties can typically be used by FSIs in three ways:
  - As part of a broader system (e.g., financial crime investigation, fraud detection, etc.) that is implemented inside the FSI.
  - As a service consumed by the FSI (e.g., contract review conducted by a professional services firm on behalf of the FSI, etc.) but using an AI model for at least part of the task.
  - As a piece of insight (e.g., an alternative credit score, a 'sentiment marker' for individual securities, etc.) that is made available to the FSI (without the underlying systems or models that helped come up with the insight)

FEAT guidelines, as well as those from other regulators, are clear in holding FSIs accountable for AIDA transparency (and fairness/ethics) in all three cases. However, in the third category, the FSI buying such a score or insight may not always have full visibility into the vendor's use of AIDA, particularly because of the vendor's concerns around sharing their IP. It may be useful to consult FSIs and providers of such third party insights (e.g., alternative credit scores) on the optimal way of addressing this challenge.

- **Definition of minimum standards around transparency.** As discussed, individual FSIs may wish to consider setting internal standards around the accuracy and consistency of transparency requirements for AIDA decision making.

  Discussions within the consortium during Phase II have exposed a natural tension between (a) FSIs' desire to retain discretion and flexibility in deciding the approach to transparency; and (ii) the benefits that FSIs see from some form of standardisation. This makes it a valuable topic for any future discussions in the consortium around transparency, particularly if/ when there is more alignment among industry forums, regulators and/or standards organisations on "acceptable" standards."

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